

IMPLEMENTING BASIC AOP (ASPECT-ORIENTED PROGRAMMING) WITH SPRING

LoggingAspect.java:

```
package com.library.aspect;

import org.aspectj.lang.JoinPoint;
import org.aspectj.lang.annotation.*;
import org.springframework.stereotype.Component;

@Aspect
@Component
public class LoggingAspect {
    @Before("execution(* com.library.service.BookService.*(..))")
    public void beforeMethod(JoinPoint joinPoint) {
        System.out.println("Before executing: " + joinPoint.getSignature().getName());
    }

    @After("execution(* com.library.service.BookService.*(..))")
    public void afterMethod(JoinPoint joinPoint) {
        System.out.println("After executing: " + joinPoint.getSignature().getName());
    }
}
```

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xmlns:context="http://www.springframework.org/schema/context"
       xmlns:aop="http://www.springframework.org/schema/aop"
       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd
           http://www.springframework.org/schema/context
           https://www.springframework.org/schema/context/spring-context.xsd
```

```
http://www.springframework.org/schema/aop
https://www.springframework.org/schema/aop/spring-aop.xsd">
```

```
<context:component-scan base-package="com.library"/>
<aop:aspectj-autoproxy/>
</beans>
```

BookRepository.java:

```
package com.library.repository;

import org.springframework.stereotype.Repository;

@Repository
public class BookRepository {
    public void display() {
        System.out.println("BookRepository method executed.");
    }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

@Service
public class BookService {
    @Autowired
    private BookRepository bookRepository;

    public void display() {
        System.out.println("BookService method executed.");
        bookRepository.display();
    }
}
```

LibraryManagementApplication.java:

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class LibraryManagementApplication {
    public static void main(String[] args) {
        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");
        BookService service = context.getBean(BookService.class);
        service.display();
    }
}
```

pom.xml:

Added dependency:

```
<dependency>
    <groupId>org.aspectj</groupId>
    <artifactId>aspectjweaver</artifactId>
    <version>1.9.22.1</version>
</dependency>
```

Output:

